

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q16-0574

New Camp 138 kV
March 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q16-0574. The Interconnection Customer proposes a 55.4 MW wind facility interconnecting at the New Camp 138 kV substation in the Kentucky area. The request was received on January 14, 2016. The study evaluated system intact and single contingency conditions using the regional planning case, including 290 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,670,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

GREENLY – DRESDEN 500 kV line: 129.5% loading (2,175 MVA) following the outage of the INTERNATIONAL PAPER – DRESDEN SUBSTATION 500 kV; DFAX = 16.0%.

For the loss of DRESDEN – INTERNATIONAL PAPER 500 kV, the OMNI-HACKBERRY – GREENLY 500 kV loads to 122.4% of its 2,579 MVA rating. Project DFAX: 15.3%.

For the loss of MANATEE – HINTON AP 345 kV, the HINTON AP 345 kV/138 kV Transformer Bank loads to 107.0% of its 520 MVA rating. Project DFAX: 5.7%.

For the loss of OMNI-HACKBERRY – GREENLY 500 kV, the DRESDEN – INTERNATIONAL PAPER 500 kV L loads to 104.6% of its 3,070 MVA rating. Project DFAX: 17.4%.

BURWELL – P.T.S 345 kV: 103.5% loading (1,246 MVA) following the outage of the ALLEGHENY ENERGY UNITS 3 4 & 5 345-1 kV XFMR; DFAX = 6.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DRESDEN – GREENLY 500 kV	Rebuild line with higher-capacity conductor	\$100,752,000	0% (\$500,000)
GREENLY – OMNI-HACKBERRY 500 kV	Rebuild line with higher-capacity conductor	\$78,445,000	0% (\$386,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
HINTON AP 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$12,951,000	1% (\$126,000)
DRESDEN – INTERNATIONAL PAPER 500 kV Line	Reconductor line	\$115,958,000	0% (\$626,000)
BURWELL – P.T.S 345 kV	Reconductor line	\$5,403,000	1% (\$32,000)
	Total allocated to project		\$1,670,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.